Morse Theory of Homotopy Type

The Cohen–Jones–Segal Construction

Derek Zeng

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Les mathématiciens n'étudient pas des objets, mais des relations entre les objets.

Mathematicians study not objects, but relations between objects.

— Henri Poincaré
La Science et l'Hypothèse, Part I, Chapter II

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Introduction

This paper is the final project for the honors topics course on Morse Theory taught by Professor Ely Kerman at the University of Illinois Urbana-Champaign (UIUC) in the fall of 2026. It surveys the Cohen–Jones–Segal approach to recovering the homotopy type of a closed smooth manifold from its critical points and gradient trajectories [CJS95b, CJS95a].

Classical Morse theory describes a manifold through handle attachments, while Morse homology records their incidence numbers by counting isolated unparametrized trajectories [Mil63, AD14]. These counts do not determine the attaching maps themselves. For example, $\mathbb{CP}^2$ and $S^2 \vee S^4$ have the same integral homology groups, but the Hopf attaching map of $\mathbb{CP}^2$ distinguishes their homotopy types and remains nontrivial after stabilization [Hat02]. This motivates studying the full trajectory spaces and their compatibility, rather than only the counts defining the differential.

Following Cohen–Jones–Segal, we review how compactified trajectory spaces assemble into a flow category, with composition given by concatenation of broken trajectories. We explain classifying-space reconstruction and the role of regularity hypotheses, drawing on Calle–Liu's treatment of tameness [CJS95b, CL26]. We then discuss the stable construction, in which compatible embeddings and stable normal framings turn trajectory spaces into stable attaching maps through the Pontryagin–Thom construction. Bonciocat's treatment supplies a rigorous account of the framing data and explains how the resulting spectra recover suspension spectra and, with suitable twists, Thom spectra [Bon24].

A broader motivation comes from Floer theory: one seeks to refine Floer homology to a spectrum that retains additional information and supports generalized homology and cohomology [CJS95a]. Finite-dimensional Morse theory provides a model for this program and clarifies the geometric structures it requires. Throughout, the emphasis is on explaining the existing constructions, distinguishing unstable from stable reconstruction, and illustrating the information lost upon passing to homology.

Conventions. Throughout this paper, without further specification, we work on connected closed Riemannian manifolds (M, g); every function f we choose is a Morse function; every pointed space is supposed to be well-pointed (i.e., (X, x) is an NDR pair).

[**Derek:** Rewrite the intro, follows the latest version of the outline]

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Main Text

1 Review of Morse-Theoretic Foundations

Elementary Facts about Morse Functions. Let (M, g) be a smooth Riemannian manifold with dimension n. Recall that a Morse function $f \in C^\infty(M)$ on M is defined as a smooth function such that all critical points are nondegenerate, i.e., for $p \in M$ such that $\mathrm{d}f_p = 0$, the Hessian can be globally defined as

$$\mathrm{Hess}^\nabla f := \nabla(\mathrm{d}f) : TM \otimes TM \to \mathbb{R} \tag{1.1}$$

where ∇ is the Levi-Civita connection. Although the Hessian clearly depends on the choice of the connection, its behavior at the critical point does not. Thus, the notion of Morse function is well-defined on any smooth manifold and does not require a Riemannian metric. More precisely, we have the following statement of well-definedness:

Proposition 1.1. *The Hessian of $f \in C^\infty(M)$ at the critical point $p \in M$ has the following properties:*

1. *The bilinear map $\mathrm{Hess}_p f : T_pM \times T_pM \to \mathbb{R}$ is independent of the choice of connection (and thus, does not depend on the metric).*
2. *The Hessian has a local coordinate representation*

$$\mathrm{Hess}_p f(v, w) = \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j} v_i w_j$$

where v_i and w_j are components of the given vectors.

Note that the second property implies the first one, and the proof is just a simple calculation of local coordinates, and one can find this argument it in [Mil63, §2, p.4-5], which we omit here.

It is a famous application of Sard's theorem in differential topology that the Morse function is generic. In fact, one can show that a Morse function always exists; furthermore, let $C^\infty_{\mathrm{Morse}}(M)$ be the collection of Morse functions on M,

Theorem 1.2. *Let M be a closed manifold, then $C^\infty_{\mathrm{Morse}}(M) \subseteq C^\infty(M)$ is a dense open subset in C^∞-topology.*

For a proof, see [Hir76, Chapter 6, §1, Theorem 1.2]. This fact generalizes to the strong C^r-topology for any $2 \leq r \leq \infty$, as stated in the reference we cite above.

Part of the reason to study Morse functions is that the Morse function has a local canonical form near critical points, which is known as the Morse lemma:

Theorem 1.3 (Morse Lemma)**.** *Let $f \in C^\infty(M)$ be a smooth function with nondegenerate critical point $p \in \mathrm{Crit}(f)$, there exists a chart (U, φ) contains p such that f has local coordinate form*

$$f \circ \varphi^{-1}(x_1, \ldots, x_n) = f(p) - x_1^2 - \cdots - x_r^2 + x_{r+1}^2 + \cdots + x_n^2. \tag{1.2}$$

The number $0 \leq r \leq n$ is called the Morse index of p, denote as $\mathrm{ind}_f(p) = r$.

The proof can be done with either some complicated linear algebra technique or by applying Moser's trick; for a complete proof, see for example [Mil63, Part I]. Another equivalent definition of Morse index includes the dimension of the negative-definite eigenspace and the number of negative eigenvalues of $\mathrm{Hess}_p f$. The classical approach to Morse theory studies the topology of a manifold through the handle attachments $D^r \times D^{n-r}$ that occur when a sublevel set crosses a critical point of index r.

Notation & Convention. For any Morse function $f \in C^\infty(M)$, we write $\mathrm{Crit}(f) \subseteq M$ for the set of critical points of f, and $\mathrm{Crit}_k(f) \subseteq \mathrm{Crit}(f)$ for the subset of critical points with Morse index equal to $k \in \mathbb{N}$.

After we established the local model underlying this picture—the Morse lemma—we can now pass directly to its homological refinement, Morse homology, using the gradient flow of Morse functions. Consider the negative gradient flow equation

$$\dot{\gamma}(s) = -\operatorname{grad} f(\gamma(s)), \tag{1.3}$$

and the let $\varphi_s : M \to M$ be the flow of Equation (1.3). Our first observation is that the critical points are hyperbolic fixed points of the flow φ_s, and the energy functional is preserved for the same endpoints:

Proposition 1.4. *Consider $p, q \in \operatorname{Crit}(f)$ with the end-points condition $\lim_{s\to-\infty}\gamma(s) = p$ and $\lim_{s\to\infty}\gamma(s) = q$. Then the energy functional of the flow line is preserved, and explicitly given by*

$$E(\gamma) := \int_{\mathbb{R}} \|\dot{\gamma}(s)\|_g^2 \mathrm{d}s = f(p) - f(q) \tag{1.4}$$

Proof. The proof is just simple computation; note that

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}s} f(\gamma(s)) &= \mathrm{d}f_{\gamma(s)}(\dot{\gamma}(s)) = \left\langle \operatorname{grad}_{\gamma(s)} f, \dot{\gamma}(s) \right\rangle \\ &= \langle -\dot{\gamma}(s), \dot{\gamma}(s)\rangle = -\|\dot{\gamma}(s)\|_g^2.\end{aligned}$$

Then, the energy functional is given by

$$E(\gamma) = \int_{\mathbb{R}} \|\dot{\gamma}(s)\|_g^2 \mathrm{d}s = -\int_{\mathbb{R}} \frac{\mathrm{d}}{\mathrm{d}s} f(\gamma(s)) \mathrm{d}s = f(p) - f(q).$$

This completes the proof. □

This is a finite-dimensional prototype of the Floer energy identity. It leads to the following properties:

1. f strictly decreases along non-constant trajectories.
2. The flow line only flows from larger critical values to smaller critical values.
3. If f takes the same value at both endpoints, i.e., $f(p) = f(q)$, then the trajectory is constant.

The classical Morse theory provides a standard way to decompose the manifold using the flow lines:

Theorem 1.5 (Fundamental Theorem of Classical Morse Theory)**.** *Let M be a (compact connected) smooth manifold and $f \in C^\infty(M)$ be a Morse function. For any value $a = f(p)$, defined the submanifold $M_a := f^{-1}(-\infty, a]$. Then, for any regular values $a < b$, the following claims can be made for M_a and M_b:*

1. *If the preimage $f^{-1}[a,b] \cap \operatorname{Crit}(f) = \varnothing$, then $M_a \cong M_b$.*
2. *If the preimage $f^{-1}[a,b] \cap \operatorname{Crit}(f) = \{p\}$ contains a single critical point with the Morse index $\operatorname{ind}_f(p) = k$, then $M_b \cong M_a \cup_\psi (D^k \times D^{n-k})$ for some attaching map $\psi : S^{k-1} \times D^{n-k} \to \partial M_a$.*

Note that all isomorphisms we write above are homeomorphisms.

For a proof of the theorem, see [AD14] or [Mil63]. The trick is to use the gradient flow of f to generate a time-dependent diffeomorphism. From the theorem above, it is straightforward that any smooth manifold is homotopy equivalent to a CW complex.

Using the gradient flow equation Equation (1.3), we may define the stable and unstable manifolds, which collect all fixed-point (critical point) information of the gradient flow:

Definition 1.6. Let $f \in C^\infty(M)$, the stable and unstable manifolds for critical point $p \in \operatorname{Crit}(f)$ are defined by the following submanifolds of M

$$\begin{aligned}W^s(p; f) &:= \left\{ x \in M \;\middle|\; \lim_{s\to\infty} \varphi_s(x) = p \right\} \\ W^u(q; f) &:= \left\{ y \in M \;\middle|\; \lim_{s\to-\infty} \varphi_s(y) = q \right\}\end{aligned}$$

Note that they are both smooth, injectively immersed submanifolds of M. More precisely,

Proposition 1.7. *Let $f \in C^\infty(M)$ be a Morse function with critical point $p \in \mathrm{Crit}(f)$. Then, the dimension of the stable and unstable manifolds is given by*

$$\dim W^u(p;f) = \mathrm{ind}_f(p), \qquad \dim W^s(p;f) = n - \mathrm{ind}_f(p).$$

Furthermore, $W^u(p;f) \cong \mathbb{R}^k$ and $W^s(p;f) \cong \mathbb{R}^{n-k}$ for the critical point with index $\mathrm{ind}_f(p) = k$.

The proof in general cases is rather complicated; however, this fact is intuitive in the following sense: Given an open neighborhood U of the critical point p with a Euclidean metric, by Morse Lemma Theorem 1.3, one can always find a local coordinate such that the Morse function has local coordinate representation of the following type

$$f(x,y) = f(p) - \frac{1}{2}\|x\|^2 + \frac{1}{2}\|y\|^2.$$

Under this choice of Euclidean local model, the gradient of f is simply $\mathrm{grad}\, f = (-x, y)$, and the gradient flow equation has solution

$$\big(x(t), y(t)\big) = (e^t x_0, e^{-t} y_0).$$

With this solution, the statement is straightforward at least locally:

1. As $t \to \infty$, the trajectory $(x,y) \to (0,0)$ if and only if $x_0 = 0$. This gives the local stable manifold $W^s_{\mathrm{loc}}(p;f) = \{(0,y) \mid y \in \mathbb{R}^{n-\lambda}\}$.
2. As $t \to -\infty$, the trajectory $(x,y) \to (0,0)$ if and only if $y_0 = 0$. This gives the local unstable manifold $W^u_{\mathrm{loc}}(p;f) = \{(x,0) \mid x \in \mathbb{R}^{\lambda}\}$.

And one can get the dimension of the stable and unstable manifolds simply by the definition of local manifolds above, and note that this also shows $T_pW^s(p;f) \cong E_p^+$ and $T_pW^u(p;f) \cong E_p^-$, where $T_pM \cong E_p^+ \oplus E_p^-$ is the splitting given by the positive and negative eigenspaces of $\mathrm{Hess}_p(f)$. The fact that the flow φ_s is a local diffeomorphism allows us to claim

$$W^s(p;f) = \bigcup_{t\geq 0} \varphi_{-t}\big(W^s_{\mathrm{loc}}(g;f)\big), \qquad W^u(p;f) = \bigcup_{t\geq 0} \varphi_t\big(W^u_{\mathrm{loc}}(g;f)\big)$$

One can try to establish a metric-invariant statement to make the method above a general claim, and we may omit any further detail of the proof here.

Morse Moduli Space. Intuitively, the intersection of stable and unstable manifolds gives a trajectory count of negative gradient flows. To ensure the intersection is indeed an embedded manifold, we may consider the following transversality condition.

Definition 1.8 (Morse-Smale Pair)**.** A pair (f,g) of smooth function f and Riemannian metric g is said to be satisfies the Morse-Smale condition if

$$W^s(p;f) \pitchfork W^u(q;f), \qquad p,q \in \mathrm{Crit}(f).$$

Such a pair of Morse function and Riemannian metric is called a Morse-Smale pair.

There is another notion of a Morse–Smale pair of pseudogradient vector fields, as we remark below:

Remark. A standard approach to Morse theory uses flows generated by pseudo-gradient vector fields, which are vector fields $X \in \Gamma(TM)$ that satisfy the following two conditions:

1. $\mathrm{d}f(X) \leq 0$, with equality only at critical points.
2. For any $p \in \mathrm{Crit}(f)$, there is a local chart $U \subseteq M$ such that $X|_U = -\mathrm{grad}\, f|_U$.

For an introduction to this method, see, for example [Mil63] or [AD14, Section 2.1].

With the transversality condition, we can define the parametrized space of the gradient flow trajectory:

Definition 1.9 (Parameterized Trajectory Space)**.** The parameterized trajectory space of the gradient flow between $p, q \in \mathrm{Crit}(f)$ is defined as $\mathcal{L}(p, q; f) := W^u(p; f) \cap W^s(q; f)$.

With some effort, we can show that this space is indeed the following submanifold: Let $W_f(p, q)$ denote the set of trajectories from p to q along $-\nabla f$

$$\mathcal{L}(p, q; f) = \{x \in M \mid \exists \gamma \in W_f(p, q) : \exists t \in (-\epsilon, \epsilon) : \gamma(t) = x\}, \tag{1.5}$$

and it dimension is given by $\dim \mathcal{L}(p, q; f) = \mathrm{ind}_f(p) - \mathrm{ind}_q(f)$. The identification in Equation (1.5) and the well-definedness of the trajectory space are due to the following proposition:

Proposition 1.10. *Let (M, g) be a compact Riemannian manifold and f be a Morse function. Then for any negative gradient trajectory $\gamma : \mathbb{R} \to M$ such that $\dot{\gamma} = -\nabla f \circ \gamma$, there exists unique $p, q \in \mathrm{Crit}(f)$ such that*

$$\lim_{t \to \infty} \gamma(t) = p, \qquad \lim_{t \to -\infty} \gamma(t) = q.$$

Proof. This is simply a compactness argument, which is basically a consequence of Proposition 1.4. We may prove the case $t \to \infty$; the other case is completely the same.

Since we are working on a compact manifold, any smooth function f is bounded, and by the monotonicity of negative gradient flow, $(f \circ \gamma)(t)$ is strictly decreasing on $\mathbb{R}$, and thus, the limit $L := \lim_{t \to \infty}(f \circ \gamma)(t)$ exists. Then we show that the limit of the trajectory is indeed a critical point. Define the limit set of accumulation points

$$\omega(\gamma) := \bigcap_{N=1}^{\infty} \overline{\gamma([N, \infty))}$$

Note that $\overline{\gamma([N, \infty))} \supseteq \overline{\gamma([N+1, \infty))} \supseteq \cdots$ are nonempty compact connected sets, we knwo that $\omega(\gamma)$ is also nonempty, compact, and connected. Now we are going to show that $\omega(\gamma) \subseteq \mathrm{Crit}(f)$. Let ϕ_t denote the flow of $-\operatorname{grad} f$; the compactness (without boundary) of M ensures that ϕ_t exists for all $t \in \mathbb{R}$, and $\gamma(s + t) = \phi_s(\gamma(t))$. Pick any $p \in \omega(\gamma)$, by the definition, one can find a sequence $t_n \to \infty$ such that $\gamma(t_n) \to p$. By continuity of f, one can show that

$$f(p) = \lim_{n \to \infty}(f \circ \gamma)(t_n) = L, \qquad \forall p \in \omega(\gamma)$$

and thus $f|_{\omega(\gamma)} \equiv L$. On the other hand, by the continuity of the flow ϕ_s,

$$\phi_s(\gamma(t)) = \lim_{n \to \infty} \phi_s(\gamma(t_n)) = \lim_{n \to \infty} \gamma(t_n + s),$$

since $t_n + s \to \infty$, the point $\varphi_s(p) \in \omega(\gamma)$. Thus, $(f \circ \phi_s) \equiv L$, and thus

$$\|\operatorname{grad} f(p)\|^2 = \mathrm{d}f_p(\operatorname{grad} f(p)) = \frac{\mathrm{d}}{\mathrm{d}s}\Big|_{s=0}(f \circ \phi_s)(p) = 0$$

which means $\operatorname{grad} f(p) = 0$ and thus $p \in \mathrm{Crit}(f)$. Note that the compactness of M requires $\mathrm{Crit}(f)$ to be finite discrete, and since $\omega(\gamma)$ is connected, $\omega(\gamma) = \{p\}$. Thus, $\gamma(t) \to p$, which completes the proof. □

Also, a trajectory has a freedom of translation, where $\mathbb{R} \curvearrowright \mathcal{L}(p, q; f)$ acts freely by

$$t \cdot \gamma(s) := \gamma(t + s).$$

What we want to study is really the orbit of this $\mathbb{R}$ action:

Definition 1.11 (Coarse Morse Moduli Space)**.** Defined the unparametrized Morse moduli space of a Morse-Smale pair (f, g) and any distinct $p, q \in \mathrm{Crit}(f)$ to be $\mathcal{M}(p, q; f) := \mathcal{L}(p, q; f)/\mathbb{R}$.

An intuitive fact is that the dimension of this trajectory space is

$$\dim \mathcal{M}(p,q;f) = \operatorname{ind}_f(p) - \operatorname{ind}_q(f) - 1.$$

One can prove this fact by considering the short exact sequence of tangent spaces induced by the inclusion of orbits. Furthermore, for distinct $p,q \in \operatorname{Crit}(f)$ and $f(q) < a < f(p)$, consider the slice given by the intersection of the Morse function level set and the Morse trajectory space

$$S_a(p,q) := \mathcal{L}(p,q) \cap f^{-1}(a)$$

The following proposition can describe the structure of the Morse moduli space:

Proposition 1.12. *Let ϕ be the negative gradient flow of the Morse function f; then it gives a diffeomorphism*

$$\Phi : S_a(p,q) \times \mathbb{R} \xrightarrow{\cong} \mathcal{L}(p,q), \qquad (x,t) \longmapsto \phi_t(x)$$

flows the intersection of a slice of the manifold with the trajectories to the trajectory space.

The proof of this proposition is simply a flow argument, which is similar to the idea of Theorem 1.5. We may give a short proof of Proposition 1.12 here and omit all the other proofs using the same technique:

Proof of Proposition 1.12. Given a point $y \in \mathcal{L}(p,q)$ where $p \neq q$. By the Picard–Lindelöf theorem on the existence and uniqueness of ODE solutions,

$$\frac{\mathrm{d}}{\mathrm{d}t}(f \circ \phi_t)(y) = -\mathrm{d}f_{\phi_t(y)}\big(\operatorname{grad} f(\phi_{\tau(y)}(y))\big) = -\|\operatorname{grad} f\big(\phi_{\tau(y)}(y)\big)\|^2 < 0$$

and in particular, $\mathrm{d}f_y(X_y) \neq 0$. Thus, $f|_{\mathcal{L}(p,q)}$ is a submersion, and $S_a(p,q) := (f|_{\mathcal{L}(p,q)})^{-1}(a)$ is a smooth hypersurface. By the strict monotonicity of f given by Proposition 1.4 and the fact that

$$\lim_{t\to-\infty} f(\phi_t(y)) = f(p) > a, \qquad \lim_{t\to\infty} f(\phi_t(y)) = f(q) < a,$$

there exists an unique $\tau(y) \in \mathbb{R}$ such that $(f \circ \phi_{\tau(y)})(y) = a$. Define $G(y,t) := (f \circ \phi_t)(y) - a$, by direct compuatation, one can obtain

$$\partial_t G\big(y, \tau(y)\big) = -\|\operatorname{grad} f\big(\phi_{\tau(y)}(y)\big)\|^2 \neq 0.$$

The implicit function theorem ensures that $\tau : \mathcal{L}(p,q) \to \mathbb{R}$ is smooth, and thus, we have a smooth inverse $\Phi^{-1}\big(\phi_t(x)\big) = \big(\phi_{\tau(y)}, -\tau(y)\big)$ of the given map $\Phi : S_a(p,q) \times \mathbb{R} \to \mathcal{L}(p,q)$. This completes the proof. □

As a corollary of Proposition 1.12, since in Definition 1.11, the moduli space is defined by the space of flow lines quotiented by the reparametrization action of $\mathbb{R}$, which acts freely on $\mathcal{L}(p,q)$, we have a regular level set model of the coarse moduli space:

Corollary 1.13. *The coarse Morse moduli space $\mathcal{M}(p,q) := \mathcal{L}(p,q)/\mathbb{R}$ can be identified as $\mathcal{M}(p,q) \cong S_a(p,q)$.*

Furthermore, by a similar flow argument to Theorem 1.5, the level set $S_a(p,q) \cong S_b(p,q)$ for $a,b \in \big(f(q), f(p)\big)$. The identification of two regular level sets corresponding to two different regular values a and b is just given by the flow that moves the level set along the trajectories:

$$S_a(p,q) \xrightarrow{\cong} S_b(p,q), \qquad z \longmapsto \phi_{\tau_b(z)}(z)$$

This model describes all unbroken trajectories.

2 Moduli Spaces and Flow Category

Compactification of Moduli Space. From another perspective, it is natural to see that we need a compactification argument on the coarse moduli space. Note that since f is strictly monotone on each (non-constant) flow line by Proposition 1.4, it defines a reparametrization of the flow line $\gamma : \mathbb{R} \to M$ with the interval $I(p,q) := (f(q), f(p))$. One can use this interval to reparametrize the flow line to the smooth curve

$$\omega_\gamma : I(p,q) \longrightarrow M, \qquad (f \circ \omega)(t) = t \quad \forall t \in \big(f(q), f(p)\big)$$

and note that this is really just a reparametrization that the decreasing rate of f is constant along the flow line, which satisfies the following ODE

$$\frac{\mathrm{d}}{\mathrm{d}t}\omega = \frac{\operatorname{grad}(f)}{\|\operatorname{grad}(f)\|^2}, \qquad \frac{\mathrm{d}}{\mathrm{d}t} f \circ \omega = \langle \operatorname{grad} f, \dot{\omega} \rangle = 1. \tag{2.1}$$

Note that each trajectory is continuous and has exactly one point at each intermediate level. Therefore, the trajectory ω can be uniquely extend as

$$\widetilde{\omega}_\gamma : \overline{I(p,q)} \longrightarrow M, \qquad (f \circ \widetilde{\omega}_\gamma)(t) = t.$$

This is a continuous trajectory $\widetilde{\omega}_\gamma \in C^0\big(\overline{I(p,q)}, M\big)$ that does not depend on the time translation of γ.

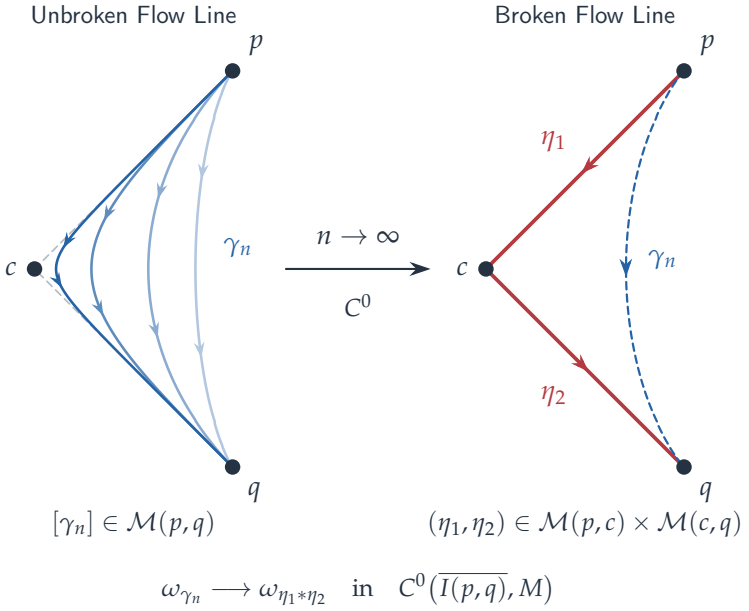
Remark. We consider the C^0-mapping space since the trajectory may not be smooth at critical endpoints.

From the idea of considering the moduli space as the solution of an **??**, we obtain a natural inclusion

$$\iota : \mathcal{M}(p,q) \hookrightarrow C^0\big(\overline{I(p,q)}, M\big).$$

However, it is intuitive that one can consider a sequence of paths $(\gamma_n) \subseteq \mathcal{M}(p,q)$ from p to q that converges to a path that is only piecewise smooth.
The idea is to construct a series of flow lines that pass through a collection of points $(x_n) \subseteq \mathcal{L}(p,q)$ such that $x_n \to c$ for some critical point $C \in \operatorname{Crit}(f)$. Thus, the flow line will converge to the concatenation of two flow lines:

$$p \xrightarrow{\eta_1} c \xrightarrow{\eta_2} q$$



which is shown in the right diagram.
This naturally leads to the idea of compactification of the Morse moduli space, which on the set level is given by considering the piecewise-smooth trajectories obtained by concatenation of flow lines in $\mathcal{M}(p,q)$, which are known as the "broken flow lines". For the convenience of labeling the broken flow lines, we introduce the poset $\mathcal{P}_f$ of the critical points:

Definition 2.1 (Poset of Critical Points)**.** The poset of critical points $\mathcal{P}_f := (\operatorname{Crit}(f), \geq)$ is defined by the collection of critical points of f together with a partial order $p \geq q$ if and only if there exists a broken flow line with finite segments that flows from $p \to q$.

We write $p > q$ if $p \geq q$ and $p \neq q$, and a descending sequence with length k from p to q is a finite sequence of critical points $\mathbf{p} := (p, p_1, \ldots, p_k, q)$ such that $p_i > p_{i+1}$, we write $|\mathbf{p}| = k$ for the length of the sequence.

Definition 2.2 (Compactified Morse Moduli Space)**.** For any critical points $p, q \in \mathrm{Crit}(f)$, define the moduli space of broken flow lines from p to q by

$$\overline{\mathcal{M}}(p,q) := \bigsqcup_{\mathbf{p}} \mathcal{M}(p,p_1) \times \cdots \times \mathcal{M}(p_k,q)$$

where the union above is a set-level disjoint union that iterates through all possible descending sequences of the critical points $\mathbf{p} = (p, p_1, \ldots, p_k, q)$ from p to q.

Remark. Note that in this definition, we need the sequence to be strictly descending, i.e., $p_i < p_j$. If not, the "inclusion" map we introduced before, $\overline{\mathcal{M}}(p,q) \hookrightarrow C^0\big(\overline{I(p,q)}, M\big)$, will not be injective, since if we allow $p_{i-1} = p_i$, then there will be three equivalent representations of the curve $p_{i-1} \to p_i$

$$p_{i-1} \xrightarrow{\gamma} p_i \xrightarrow{\mathrm{id}_i} p_i \ , \qquad p_{i-1} \xrightarrow{\gamma} p_i \ , \qquad p_{i-1} \xrightarrow{\mathrm{id}_{i-1}} p_{i-1} \xrightarrow{\eta} p_i$$

which will be sent to the same curve in the inclusion.

Remark. In the following passage, we do not distinguish the image of the moduli space $\overline{\mathcal{M}}(p,q)$ under ι and the compactified moduli space itself.

The following special cases of Morse moduli spaces are useful to remember:

1. By convention, when $p = q$, we require the moduli space $\overline{\mathcal{M}}(p,p) = \{\mathrm{id}_p\}$ to be a single point.
2. When $\mathrm{ind}_f(p) = \mathrm{ind}(q) + 1$, the moduli space $\overline{\mathcal{M}}(p,p) = \mathcal{M}(p,q)$ is discrete and compact.

And the boundary strata of $\overline{\mathcal{M}(p,q)}$ refers to the space

$$\overline{\mathcal{M}(p,q)} \setminus \mathcal{M}(p,q) = \bigsqcup_{|\mathbf{p}|\geq 1} \mathcal{M}(p,p_1) \times \cdots \times \mathcal{M}(p_k,q)$$

containing all broken flow lines of the compactified moduli space.

Remark. Note that it is clear that this is not a coproduct in Top; one should not give the compactified Morse moduli space the coproduct topology. Instead, it inherits a natural topology from the mapping space $C^0\big(\overline{I(p,q)}, M\big)$, which respects the convergence of trajectories under the metric

$$\mathrm{dist}_\infty(\gamma_1,\gamma_2) := \sup_{t\in\overline{I(p,q)}} \mathrm{dist}_M\big(\gamma_1(t),\gamma_2(t)\big), \qquad \forall \gamma_1,\gamma_2 \in C^0\big(\overline{I(p,q)}, M\big).$$

This metric is independent of parametrization.

We may show that this is indeed a compactification of the Morse moduli space:

Theorem 2.3 (Compactness)**.** *The moduli space $\overline{\mathcal{M}(p,q)}$ is a compact metrizable space.*

To prove this compactness theorem, we need some technical lemmas: Let $V_f := f\big(\mathrm{Crit}(f)\big)$ be the collection of regular values of the Morse function f,

Lemma 2.4 (Gradient Estimation)**.** *For every $x \in M$, there is constant $A > 0$ which only depends on the Morse–Smale pair (f,g) such that*

$$\|\operatorname{grad} f(x)\|_g \geq A\sqrt{\mathrm{dist}_{\mathbb{R}}\big(f(x), V_f\big)}. \tag{2.2}$$

Proof. Fix a critical point $r \in \mathrm{Crit}(f)$ and a local coordinate (U_r, z) near the critical point r. By Taylor's expansion in local coordinates, one can write

$$(\operatorname{grad} f)_i(z) = \sum_{j,k} g^{ij} H_{jk}(z) z_k + O(\|z\|^2), \qquad f(z) - f(r) = \frac{1}{2}\sum_{i,j} H_{ij}(z) z_i z_j + O(\|z\|^3),$$

where $H_{ij} = \mathrm{Hess}(f)_{ij}$ is the Hessian. Since f is Morse, H_{ij} is invertible, and one can consider the uniform estimation of the Riemannian and the Euclidean norm; one can find $\alpha_r, \beta_r > 0$

$$\|\operatorname{grad}(f)\|_g \geq \alpha_r \|z\|_{\mathbb{R}^n}, \qquad |f(z) - f(r)| \leq \beta_r \|z\|^2_{\mathbb{R}^n}.$$

Then, one may apply these inequalities and obtain the estimate

$$\|\operatorname{grad}(f)\|_g \geq \alpha_r \|z\|_{\mathbb{R}^n} \geq \frac{\alpha_r}{\sqrt{\beta_r}} \sqrt{|f(z) - f(r)|} \geq \frac{\alpha_r}{\sqrt{\beta_r}} \sqrt{\operatorname{dist}_{\mathbb{R}}\left(f(z, V_f)\right)}.$$

On the other hand, consider the compact space

$$K := M \setminus \bigcup_{r \in \mathrm{Crit}(f)} U_r$$

without any critical points of f. So if $K \neq \varnothing$, one can pick

$$m := \min_{x \in K} \|\operatorname{grad} f(x)\|_g > 0$$

and this argument together with the coefficient on a finite number of neighborhoods gives a global coefficient A. This completes the proof of the estimation. □

This estimation gives a Hölder bound for the flow lines and leads to the equicontinuity of the flow line.

Lemma 2.5 (Uniform 1/2-Hölder Estimation)**.** *There exists number $C > 0$, such that any flow trajectory $\omega \in \overline{\mathcal{M}(p,q)}$ satisfies*

$$\operatorname{dist}_M\left(\omega(s), \omega(t)\right) \leq C|t-s|^{1/2}, \tag{2.3}$$

and furthermore, this family of curves is equicontinuous.

Proof. In the interior of each non-constant flow trajectory, by the normalized gradient equation Equation (2.1) and Lemma 2.4, for any noncritical value $t \notin V_f$

$$\|\dot{\omega}(t)\| = \frac{1}{\|\operatorname{grad} f(\omega(t))\|} \leq \frac{1}{A\sqrt{\operatorname{dist}_{\mathbb{R}}\left(t, V_f\right)}}$$

Given $s < t$, we partition the interval $[s,t]$ at a finite number of critical values and integrate over each subinterval; then, using the continuity of the curve at the dividing endpoints and the triangle inequality, we can obtain the following estimation:

$$\begin{aligned}\operatorname{dist}_M\left(\omega(s), \omega(t)\right) &\leq \frac{1}{A}\int_s^t \frac{\mathrm{d}u}{\sqrt{\operatorname{dist}_{\mathbb{R}}\left(u, V_f\right)}} = \frac{1}{A}\int_s^t \max_{c_i \in V_f} \frac{1}{\sqrt{|u-c_i|}}\mathrm{d}u \\ &\leq \frac{1}{A}\sum_j \int_s^t \frac{\mathrm{d}u}{\sqrt{|u-c_i|}}.\end{aligned}$$

Note that for $c \in [s,t]$, one can compute the integration

$$\int_s^t \frac{\mathrm{d}u}{\sqrt{|u-c_i|}} = 2\sqrt{c_i - s} + 2\sqrt{t - c_i} \leq 2\sqrt{2}\sqrt{t-s}$$

and for $c \neq [s,t]$, write $d = \operatorname{dist}_{\mathbb{R}}(c, [s,t])$ and $h = t - s$

$$\int_s^t \frac{\mathrm{d}u}{\sqrt{u-c_i}} = 2(\sqrt{d+h} - \sqrt{d}) \leq 2\sqrt{2}\sqrt{t-s}.$$

Thus, suppose $N = |\operatorname{Crit}(f)|$, then

$$\operatorname{dist}_M\big(\omega(s),\omega(t)\big) \le \frac{2\sqrt{2}N}{A}\sqrt{t-s}$$

This completes the proof of the lemma. □

Write $a = f(p)$ and $b = f(q)$ for $p, q \in \operatorname{Crit}(f)$, suppose our trajectory flows from p to q. Now we may prove the compactness theorem:

Proof of Theorem 2.3. To prove the theorem, we need to show that any sequence of curves (ω_n) in the moduli space $\overline{\mathcal{M}}(p,q)$ admits a uniformly convergent subsequence with limit ω, and the limit is indeed a (possibly broken) flow line.

Lemma 2.5 allows us to apply the Arzelà–Ascoli theorem (see any standard functional analysis textbook, for example [Con90, Theorem VI.3.8]) to obtain a sequential compactness argument. By the Arzelà–Ascoli theorem, we can implicitly claim the existence of a (uniformly, since we are working on the C^0-topology) convergent subsequence in the mapping space $C^0([b,a], M)$, which we still denote, by abuse of notation, as $\omega_n \to \omega$, with the limit satisfying the following conditions:

$$(f \circ \omega)(u) = u, \qquad \omega(a) = p, \qquad \omega(b) = q.$$

To show that this is still a broken trajectory, consider the smooth vector field on $M \setminus \operatorname{Crit}(f)$ defined by

$$Y := \operatorname{grad} f / \|\operatorname{grad} f\|_g^2.$$

We first prove that for any $\tau_0 \in [a,b]$ such that $\omega(\tau_0) \notin \operatorname{Crit}(f)$, there exists a neighborhood J of u_0 that is disjoint from the critical points such that the normalized gradient equation (2.1) is satisfied

$$\dot{\omega}(\tau) = Y\big(\omega(\tau)\big)$$

in some neighborhood of τ_0. Pick a coordinate chart (U, y) that contains no critical points, write $y_n = y(\omega(t))$ as the coordinate representation of the flow lines in the sequence and let Y_U be the local coordinate representation of Y. By the continuity of the flow line, one can find a sufficiently small closed interval J containing τ_0 such that $\omega(J) \subseteq K \subseteq U$ for some compact set K. The uniform convergence ensures that for sufficiently large n, $\omega_n(J) \subset U$, and thus, one can assume ω_n has no breaking in U and satisfies Eq.(2.1). Then, in local coordinates,

$$y_n(t) - y_n(s) = \int_s^t Y_U\big(y_n(\tau)\big)\mathrm{d}\tau \xrightarrow[\text{unif. cov.}]{n\to\infty} y(t) - y(s) = \int_s^t Y_U\big(y(\tau)\big)\mathrm{d}\tau.$$

Thus y is C^1, and satisfies $\dot{y} = Y_U(y)$. By the smoothness of Y_U, we can obtain the smoothness of $y(t)$. This ensures that the flow line

$$y(t) = y(s) + \int_s^t Y_U\big(y(\tau)\big)\mathrm{d}\tau$$

satisfies the Eq.(2.1).

Let $B = \{u \in [b,a] \mid \omega(u) \in \operatorname{Crit}(f)\}$, since $f(\omega(u)) = u$, we have the fact that $B \subseteq V_f \cap [b,a]$, and thus B is a finite set that contains a, b. We can write

$$b = c_0 < c_1 < \cdots < c_\ell = a, \qquad r_i = \omega(c_i) \in \operatorname{Crit}(f).$$

On each open interval (c_{i-1}, c_i), let ω is an integral curve of Y. We need to transform this back into negative-gradient time. Fix $\tau_* \in (c_{i-1}, c_i)$, defined

$$t(\tau) = -\int_{t_*}^t \frac{\mathrm{d}s}{\big\|\operatorname{grad} f\big(\omega(s)\big)\big\|^2}.$$

Since $t'(\tau) < 0$ is strictly monotonically decreasing, this reparametrization has inverse $\tau(t)$, and let

$$\gamma_i(t) := (\omega \circ \tau)(t), \qquad \frac{\mathrm{d}\gamma_i}{\mathrm{d}t} = \frac{\operatorname{grad} f\big(\omega(\tau)\big)}{\big\|\operatorname{grad} f\big(\omega(\tau)\big)\big\|^2}\frac{\mathrm{d}\tau}{\mathrm{d}t} = -\operatorname{grad} f\big(\gamma_i(t)\big).$$

It should also be noted that the range of this new parameter is the entire set $\mathbb{R}$. Suppose a certain critical endpoint corresponds to a finite t. Then γ_i can be extended up to this finite time, reaching a critical point at that instant. By the uniqueness property of smooth ODEs, any solution passing through a critical point must be a constant solution; however, this contradicts the fact that γ_i is nonconstant. Therefore,

$$\lim_{t\to-\infty}\gamma_i(t) = r_i, \qquad \lim_{t\to\infty}\gamma_i(t) = r_{i-1}.$$

These trajectories constitute a broken flow line in the form of

$$p = r_\ell \longrightarrow r_{\ell-1} \longrightarrow \cdots \longrightarrow r_0 = q.$$

Thus, we obtain the limit $\omega_n \to \omega \in \overline{\mathcal{M}}(p,q)$. We prove that this space is sequentially compact. Since it is a subspace of a uniform metric space, sequential compactness is equivalent to compactness. This completes the proof of the compactness theorem. □

The Morse moduli space is actually a more general type of space called a manifold with corners, which we will briefly introduce here. This object is strongly related to the study of singular spaces, especially the moduli problem of J-holomorphic curves in symplectic topology. For more detailed study of manifolds with corners and related topics, see [Joy12, Lau00].

Recall that a manifold with boundary is a manifold with the local model given by a maximal smooth atlas of charts (U, φ), where $U \subseteq M$ is an open set and

$$\varphi : U \xrightarrow{\cong} \varphi(U) \subseteq \mathbb{H}^n, \qquad \mathbb{H}^n := \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid x_1 \geq 0\}.$$

As a generalization of the manifold with boundary, we may consider a manifold with more complicated boundary strata, namely, a manifold with corners, which is defined as follows:

Definition 2.6 (Manifold with Corners, [Joy12, Definition 2.1])**.** A n-dimensional manifold with corners is a Hausdorff and second countable topological space with local model given by a maximal smooth atlas of charts (U, φ), where $U \subseteq M$ is some open set and

$$\varphi : U \xrightarrow{\cong} V, \qquad V \subseteq [0,\infty)^k \times \mathbb{R}^{n-k}, \qquad 0 \leq k \leq n$$

be the coordinate function where $V \subseteq [0,\infty)^k \times \mathbb{R}^{n-k}$ be some relative open set.

Furthermore, since in this definition, we do not require k to be a fixed number for each chart, we can define the depth of a point to show the boundary property:

Definition 2.7 (Depth, [Joy12, Definition 2.3])**.** For point $p \in M$, the depth of p is the number of coordinates in $\varphi(p)$ that are zero for some chart (U, φ) such that $p \in U$.

One can check the well-definedness of this notation:

Proposition 2.8. *The depth of corner points is well-defined, i.e., it does not depend on the choice of charts.*

The proof is simply the same as the well-definedness of the boundary points on any manifold with boundary which is just a technical application of the inverse function theorem, and we leave this proof as an exercise for the reader to show. For smooth n-dimensional manifold with corner M, we write

$$S^k(M) := \{x \in M \mid \operatorname{depth}_M(z) = k\}.$$

Then, we have the stratification of the manifold with corner:

1. $S^k(M)$ is an $(n-k)$-dimensional smooth manifold without boundary (can be empty).
2. There is a decomposition
$$M = \bigsqcup_{k=0}^{n} S^k(M).$$
3. There is a closure relation
$$\overline{S^k(M)} = \bigcup_{l \le k} S^l(M).$$

For a proof of properties of manifolds with corners [Joy12, Proposition 2.4]. Under the standard assumption of the Euclidean Morse–Smale pair, such that for any local coordinate near a critical point $p \in \mathrm{Crit}(f)$, there is a Morse lemma chart with coordinates $(x, y) \in \mathbb{R}^\lambda \times \mathbb{R}^{n-\lambda}$ for $\lambda := \mathrm{ind}_f(p)$ such that

$$f = f(p) - \frac{1}{2}\|x\|_g^2 + \frac{1}{2}\|y\|_g^2, \qquad g = \sum_i \mathrm{d}x_i \otimes \mathrm{d}x_i + \sum_j \mathrm{d}y_j \otimes \mathrm{d}y_j$$

Under this assumption (which clearly does not really affect the topological result), we claim further about the structure of compactified Morse moduli space that

Theorem 2.9. *Let M be a closed Riemannian n-manifold and (f, g) be a Morse–Smale pair. For distinct critical points $p, q \in \mathrm{Crit}(f)$, the compacticifed Morse moduli space $\overline{\mathcal{M}}(p,q) \subseteq C^0(\overline{I(p,q)}, M)$ is smooth compact manifold with corners under C^0-topology with dimension*

$$\dim \overline{\mathcal{M}}(p,q) = \mathrm{ind}_f(p) - \mathrm{ind}_f(q) - 1.$$

In particular, the moduli space satisfies:

1. *$\overline{\mathcal{M}}(p,q)$ is compact.*
2. *The interior of the compactified moduli space $\mathrm{int}\big(\overline{\mathcal{M}}(p,q)\big) = \mathcal{M}(p,q)$ is the uncompactified moduli space. Moreover, the smooth structure on the interior coincides with the original smooth structure of the uncompactified Morse moduli space.*
3. *The depth-k stratum of $\overline{\mathcal{M}}(p,q)$ are given by the broken flow lines with k-breakings:*
$$S^k\overline{\mathcal{M}}(p,q) = \bigsqcup_{p=r_0>r_1>\cdots>r_{k+1}=q} \mathcal{M}(p,r_1) \times \mathcal{M}(r_1,r_2) \times \cdots \times \mathcal{M}(r_k,q).$$

A stronger version of the theorem claims the existence of a global contraction that models the compactified Morse moduli space by identifying it with a region of the manifold M (namely, global charts); for more detail, see [Weh12, Theorem 2.5].

To prove this theorem, we need to introduce the notion of gluing on paths. The gluing homeomorphism states that for any broken trajectory in the moduli space, one can find a neighborhood of unbroken trajectories, and this gluing homeomorphism directly gives the corner charts on the compactified moduli space $\overline{\mathcal{M}}(p,q)$. In this way, we can prove Theorem 2.9.

The Gluing Argument. Give a k-fold broken trajectory $(\gamma_0, \ldots, \gamma_k)$ and let the dimension of the underlying manifold $\dim M = n$, this is a point in the stratum with dimension

$$\sum_{i=0}^{k} \big(\mathrm{ind}_f(r_i) - \mathrm{ind}_f(r_{i+1}) - 1\big) = \mathrm{ind}_f(r_i) - \mathrm{ind}_f(r_{i+1}) - k + 1 = \dim \overline{M}(p,q) - k.$$

We expect to find a local model in the form $U \times [0,\delta)^k$ for some open set $U \subseteq \mathbb{R}^{n-k}$. The construction of such corner charts is exactly the gluing argument, which we need to prove to find a nearby unbroken trajectory for any broken trajectory. More precisely, we need the following gluing theorem:

Theorem 2.10 (Gluing and Smoothness). *Let M be a closed smooth manifold, and let (f, g) be a locally Euclidean Morse–Smale pair as stated. For any k-fold broken trajectory $\gamma := (\gamma_0, \dots, \gamma_k)$ flows from p to q in the form*

$$p = r_0 \longrightarrow r_1 \longrightarrow \cdots \longrightarrow r_k \longrightarrow r_{k+1} = q.$$

There exists a neighborhood $\gamma_i \in U_i \subset \overline{\mathcal{M}}(r_i, r_{i+1})$ and $\delta > 0$, such that there exists homeomorphism:

$$\mathrm{Gl}_\gamma : \left(\prod_{i=1}^{k} U_i\right) \times [0, \delta)^k \xrightarrow{\cong} \mathcal{V} \subset \overline{\mathcal{M}}(p, q)$$

where $\mathcal{V}$ is a neighborhood of γ and these data satisfies:

1. *When all parameters are zero, $\mathrm{Gl}_\gamma(\eta_0, \dots, \eta_k, 0, \dots, 0) = (\eta_0, \dots, \eta_k)$.*
2. *For any nonzero parameters $t_i > 0$, $\mathrm{Gl}_\gamma(\eta_1, \dots, \eta_k, t_1, \dots, t_k)$ gives a unbroken flow line.*
3. *For different broken flow lines, the corresponding charts have smooth transition maps.*

This gluing theorem leads to the structure on the compactified Morse moduli space as a manifold with corner, which implies Theorem 2.9.

[**Derek:** To be complete.]

Flow Category. From the compactified Morse Moduli spaces and the gluing argument, it is natural to think about the following construction of a category, namely, the flow category.

Notation & Convention. For any category $\mathcal{C}$, we denote in short the set of morphisms between any two objects $X, Y \in \mathrm{ob}(\mathcal{C})$ as $\mathcal{C}(X, Y) := \mathrm{Hom}_{\mathcal{C}}(X, Y)$.

Definition 2.11 (Flow Category). Let (M, g) be a closed Riemannian manifold, and let $f : M \to \mathbb{R}$ be a Morse function. The flow category $\mathcal{C}_f$ of f is defined as a Top-enriched category with the following data:

1. The object $\mathrm{ob}\,\mathcal{C}_f = \mathrm{Crit}(f)$ be all critical points of f.
2. For any two critical points $p, q \in \mathrm{ob}(\mathcal{C}_f) = \mathrm{Crit}(f)$, the set of morphisms is defined to be the compactified Morse moduli space:

$$\mathcal{C}_f(p, q) := \overline{\mathcal{M}}(p, q) = \bigsqcup_{\mathbf{p}} \mathcal{M}(p, p_1) \times \cdots \times \mathcal{M}(p_k, q)$$

3. The composition of morphisms [**Derek:** To be complete.]

It is clear that this category is "small" in the sense of a Grothendieck universe, so we can ignore any set-theoretic problems.

3 Unstable Cohen–Jones–Segal Realization

Definition 3.1 (Tame Morse Functiuon, [CL26, Definition 2.12.]). A Morse function $f : M \to \mathbb{R}$ is tame if $\overline{\mathcal{M}}(p, q)$ is locally contractible for all $p, q \in \mathrm{Crit}(f)$.

[**Derek:** Specify the convention for local contractibility if it has not already been introduced: for every point x and open neighborhood V of x, there is an open neighborhood $U \subseteq V$ of x such that $U \hookrightarrow V$ is null-homotopic. This is the convention in [CL26, Definition 2.10].]

If f is a tame Morse function on M, then the following Cohen–Jones–Segal theorem holds:

Theorem 3.2 (Cohen–Jones–Segal, [CJS95b, Theorem 6.1], [CL26, Theorem 3.1]). *Let M be a closed Riemannian manifold and $f : M \to \mathbb{R}$ be a tame Morse function. Then there is a homotopy equivalence $\mathbf{B}\mathcal{C}_f \simeq M$.*

Remark. The original statement of this theorem in [CJS95b, Theorem 6.1] claims that for any Morse function f on M, we have such a homotopy equivalence. However, as [CL26, Section 4] stated, this result needs either the Morse–Smale condition or the tameness condition.

[**Derek:** Explain why tameness implies that the compactified moduli spaces have CW homotopy type, and deduce that $\mathbf{B}\mathcal{C}_f$ has CW homotopy type. This does not assert that the index filtration is a CW skeletal filtration.]

Proof of Theorem 3.2. [**Derek:** Introduce the auxiliary comparison space $\mathbf{B}\widetilde{\mathcal{C}}_f$ and establish $\mathbf{B}\widetilde{\mathcal{C}}_f \simeq \mathbf{B}\mathcal{C}_f$.]

[**Derek:** Following [CL26, Section 3.3], construct the evaluation map $\mathbf{B}\widetilde{\mathcal{C}}_f \to M$ and verify the hypotheses of Smale's mapping theorem to obtain a weak homotopy equivalence.]

[**Derek:** Apply Whitehead's theorem using the CW homotopy types established above.] □

Our main theorem, Theorem 3.2, can be greatly improved under the Morse–Smale condition on (f, g):

Theorem 3.3 (Cohen–Jones–Segal)**.** *If (f, g) satisfies the Morse–Smale condition of transversality, then there is a homeomorphism $\mathbf{B}\mathcal{C}_f \cong M$.*

For a proof, see [CJS95b, Section 2-3]. The proof uses technical details of the construction of the geometric realization on $\mathbf{B}\mathcal{C}_f = |\mathrm{N}_\bullet \mathcal{C}_f|$ as a combinatorial space and the time-dependent gluing construction on $\overline{\mathcal{M}}(p, q)$, which do not fit with the main idea of this paper[3.1], and we will not show them here.

4 Stable Cohen–Jones–Segal Realization

[**Derek:** Include the following contents:

- Begin with the Morse flow category $\mathcal{C}_f$ constructed in Section 2 and the reconstruction $B\mathcal{C}_f \simeq M$ from Section 3. Explain the goal of constructing a spectrum directly from trajectory spaces through Pontryagin–Thom attaching maps. Throughout this section, work with a Morse–Smale pair (f, g) on the closed manifold M, with standard local Morse data near the critical points. Retain the notation $\overline{\mathcal{M}}(p, q)$ and use $\mathrm{ind}_f(p)$ as the homological grading.
- Recall the smooth geometric input needed from the gluing argument in Section 2:

  $$\dim \overline{\mathcal{M}}(p, q) = \mathrm{ind}_f(p) - \mathrm{ind}_f(q) - 1, \qquad \partial_r \overline{\mathcal{M}}(p, q) \cong \overline{\mathcal{M}}(p, r) \times \overline{\mathcal{M}}(r, q).$$

  Specify compatible smooth corner structures and collars, including their compatibility under iterated breaking. Cite the precise smooth compactification result being used; the compactness theorem and tameness condition established earlier do not themselves supply this additional structure.
- Introduce compatible neat embeddings, tubular neighborhoods, and stable normal framings for these compactified trajectory spaces. Explain their product compatibility along boundary faces. Following Bonciocat, introduce the compactified spaces $\overline{W^u(p; f)}$ of trajectories with free endpoint and their evaluation maps to M; explain their role in constructing the standard untwisted framing system. State the framing existence result and describe its geometric mechanism, including compatibility at the corners. No stable parallelizability assumption on M is required.
- Explain the relative Pontryagin–Thom construction for the framed trajectory spaces and use it to construct a based CW complex $X_N(f)$ with one non-basepoint cell of dimension $N + \mathrm{ind}_f(p)$ for each $p \in \mathrm{Crit}(f)$. Describe explicitly how the collapse maps determine the attaching maps. Include a commutative diagram expressing compatibility with a boundary product $\overline{\mathcal{M}}(p, r) \times \overline{\mathcal{M}}(r, q)$, and explain why iterated compatibility makes the attachments well-defined at higher corners.
- Define the normalized stable realization

  $$\mathbb{X}_f := \Sigma^{-N} \Sigma^\infty X_N(f).$$

  Explain the dimension shift N and show that compatible stabilization of the auxiliary embedding data gives

  $$X_{N+r}(f) \simeq \Sigma^r X_N(f),$$

[3.1] This homeomorphism construction does not generalize to Floer theory.

leaving $\mathbb{X}_f$ unchanged up to stable equivalence. State invariance under homotopies of the compatible geometric data. Keep the framing system fixed in this statement; dependence on other framing systems will be examined in Section 6.2.

- State and explain the stable Morse reconstruction theorem: with the standard untwisted Morse framing data,
$$\epsilon_f : \mathbb{X}_f \xrightarrow{\simeq} \Sigma^\infty M_+.$$
Use the free-endpoint trajectory spaces to construct the comparison map. Outline the proof by comparing the filtration by critical values with the Morse handle filtration, identifying the induced maps on the successive quotients, and applying induction through cofiber sequences. Relate the conclusion to the classifying-space reconstruction of Section 3. Use Bonciocat's Theorem 5.9, specialized to the untwisted Morse case, as the main reference.
- Prepare the transition to Section 5 by retaining the index filtration. For the full subcategory $\mathcal{C}_f^k$ on critical points of index at most k, let $\mathbb{X}_f^k$ denote its realization using the restricted geometric data. Establish
$$\operatorname{cofib}\left(\mathbb{X}_f^{k-1} \longrightarrow \mathbb{X}_f^k\right) \simeq \bigvee_{p \in \operatorname{Crit}_k(f)} \Sigma^k \mathbb{S}, \qquad \mathbb{X}_f^{-1} = 0.$$
Explain briefly that the incidence numbers are the signed counts of zero-dimensional trajectory spaces. Leave the systematic identification with $CM_\bullet(f,g)$, generalized coefficients, and the Atiyah–Hirzebruch spectral sequence to Section 5. The comparison with its filtration on $B\mathcal{C}_f$ must be justified by a filtered argument.
- Close by identifying the outputs needed later: the spectrum $\mathbb{X}_f$, its stable attaching maps, and its index filtration. These support the $\mathbb{CP}^2$ calculation and twisted-framing examples in Section 6. Reserve the abstract realization theorem, Floer input, and continuation formalism for Section 7. Use Bonciocat, *Revisiting the Cohen–Jones–Segal construction in Morse–Bott theory* (arXiv:2409.11278v3), Sections 4–5, as the principal reference, specialized throughout to isolated critical points and the untwisted construction.

]

5 Generalized Morse Homology

Review of Morse Homology. We first recall classical Morse homology and its interpretation through the index filtration of the Morse flow category. We then use the stable CJS realization to extend this picture to generalized homology theories.

[**Derek:** Review Morse Homology.]

If the pair (f,g) satisfies the Morse–Smale condition, using what we construct in Section 3, there is a clean construction of the index spectral sequence on the classifying space of the Morse flow category.

Theorem 5.1 (Index Spectral Sequence, [CJS95b, Theorem 4.1])**.** *Consider the index filtration on the classifying space of the Morse flow category $\mathcal{C}_f$ induced by the Morse index*

$$\mathbf{B}\mathcal{C}_f^0 \subseteq \mathbf{B}\mathcal{C}_f^1 \subseteq \cdots \mathbf{B}\mathcal{C}_f^k \subseteq \mathbf{B}\mathcal{C}_f^{k+1} \subseteq \cdots \subseteq \mathbf{B}\mathcal{C}_f \simeq M$$

where $\mathcal{C}_f^k$ is the full subcategory that contains all critical points with index less than or equal to k. Then, for a Morse–Smale pair (f,g), one can define the index spectral sequence, and the following claims hold

1. *There is a homotopy equivalence*
$$\mathbf{B}\mathcal{C}_f^k / \mathbf{B}\mathcal{C}_f^{k-1} \simeq \bigvee_{p \in \operatorname{Crit}_k(f)} S_p^k$$

2. *In the index spectral sequence,*

$$E_1^{r,s} = H_{r+s}(\mathbf{B}\mathcal{C}_f^r, \mathbf{B}\mathcal{C}_f^{r-1}) \cong \begin{cases} \bigoplus_{p\in \mathrm{Crit}_r(f)} \mathbb{Z}[p], & \text{if } s = 0 \\ 0, & \text{otherwise.} \end{cases}$$

and thus, the E_1-page is given by the Morse chain compelx $\mathrm{CM}_\bullet(f,g)$

$$\mathrm{CM}_\bullet(f,g) \cong \left[\cdots \longrightarrow E_1^{r+1,0} \longrightarrow E_1^{r,0} \longrightarrow E_1^{r-1,0} \longrightarrow \cdots \right]$$

This spectral sequence corresponds to the Atiyah–Hirzebruch spectral sequence of the CW filtration.

Proof of Theorem 5.1. [**Derek:** Under the Morse–Smale hypothesis, establish the filtration quotients independently of the unfiltered homotopy equivalence in Theorem 3.2. Explain the required attaching-space or filtered comparison argument.]

[**Derek:** Deduce the description of the E_1-page from the relative homology groups. Treat filtration degree zero separately, or specify the based quotient convention.]

[**Derek:** Choose orientations of the unstable manifolds and identify d_1 with the signed count of trajectories. This identifies $(E_1^{\bullet,0}, d_1)$ with $\mathrm{CM}_\bullet(f,g)$ as chain complexes.] □

[**Derek:** Explain the filtered comparison with the Morse CW filtration. This identifies the spectral sequence with the ordinary-homology case of the Atiyah–Hirzebruch spectral sequence; an equivalence of the total spaces alone does not establish this identification.]

A straightforward corollary is

Corollary 5.2. *The index spectral sequence collapses to the homology of the manifold $H_\bullet(M) \cong \mathrm{HM}_\bullet(f,g)$ on the E_2-page. More precisely,*

$$E_2^{r,s} \cong \begin{cases} H_r(M), & \text{if } s = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Spectrum Coefficients and Generalized Morse Homology. [**Derek:** Include the following contents:

- Let E be a spectrum and define

$$\mathrm{HM}_k(f,g;E) := \pi_k(E \wedge \mathbb{X}_f),$$

  where $\mathbb{X}_f$ is the untwisted stable Morse realization constructed in Section 4 and all smash products are derived. Explain that this definition uses the entire spectrum, including its stable attaching maps. No ring structure on E is needed at this stage.
- Apply the stable reconstruction theorem to obtain

$$\mathrm{HM}_k(f,g;E) \cong \pi_k(E \wedge \Sigma^\infty M_+) = E_k(M).$$

  State the grading convention and explain independence of the Morse data at the level of these groups. Give $E = H\mathbb{Z}$ and $E = \mathbb{S}$ as the basic examples, recovering ordinary homology and stable homotopy groups, respectively. Refer to Appendix B for the required facts about spectra and generalized homology.

]

Stable Index Filtration and Atiyah–Hirzebruch Spectral Sequence. [**Derek:** Include the following contents:

- Use the index filtration from Section 4,

$$0 = \mathbb{X}_f^{-1} \longrightarrow \mathbb{X}_f^0 \longrightarrow \cdots \longrightarrow \mathbb{X}_f^n = \mathbb{X}_f, \qquad n = \dim M.$$

Since smashing with E preserves cofiber sequences, its successive quotients are

$$\operatorname{cofib}\left(E \wedge \mathbb{X}_f^{r-1} \longrightarrow E \wedge \mathbb{X}_f^{r}\right) \simeq \bigvee_{p \in \mathrm{Crit}_r(f)} \Sigma^r E.$$

Construct the resulting exact couple, displaying its diagram, and obtain

$$E^1_{r,s}(f;E) \cong \bigoplus_{p \in \mathrm{Crit}_r(f)} \pi_s(E).$$

Fix the homological differential convention

$$d^\ell : E^\ell_{r,s} \longrightarrow E^\ell_{r-\ell,s+\ell-1}.$$

- Identify d^1 through the Pontryagin–Thom maps of the zero-dimensional trajectory spaces. With compatible orientation and suspension conventions, show that

$$(E^1_{\bullet,s}, d^1) \cong CM_\bullet(f,g) \otimes_{\mathbb{Z}} \pi_s(E),$$

and therefore

$$E^2_{r,s}(f;E) \cong H_r(M;\pi_s(E)).$$

Explain that these are constant coefficients in the untwisted setting considered here.

- Establish convergence

$$E^2_{r,s}(f;E) \Longrightarrow \mathrm{HM}_{r+s}(f,g;E) \cong E_{r+s}(M).$$

Define the filtration on the abutment by

$$F_r HM_k(f,g;E) := \operatorname{im}\left(\pi_k(E \wedge \mathbb{X}_f^r) \longrightarrow HM_k(f,g;E)\right),$$

so that

$$E^\infty_{r,s} \cong F_r HM_{r+s}(f,g;E) \,/\, F_{r-1}\,\mathrm{HM}_{r+s}(f,g;E).$$

Explain why the finite index filtration gives strong convergence, even when E is nonconnective, and why extension problems may remain. Identify this as the cellular Atiyah–Hirzebruch spectral sequence of the CJS realization. To compare it with a chosen Morse CW filtration of M, supply the filtered comparison; the equivalence of total spectra alone does not identify spectral sequences.

]

Recovering the Ordinary Morse Complex. [**Derek:** Include the following contents:

- Specialize to $E = H\mathbb{Z}$. Since $\pi_s(H\mathbb{Z}) = 0$ for $s \neq 0$, the E^1-page is concentrated in one row, with differential

$$\partial p = \sum_{\mathrm{ind}_f(q) = \mathrm{ind}_f(p)-1} \#\mathcal{M}(p,q)\, q.$$

Recover the ordinary Morse complex and its homology, connecting this calculation with Theorem 5.1 and Corollary 5.2.

- Explain the difference for general E: several coefficient rows can occur, and higher differentials can retain information from stable attaching maps that the ordinary Morse differential does not detect. Consequently, the homology of $CM_\bullet(f,g) \otimes \pi_*(E)$ generally describes the E^2-page rather than the final generalized homology. This supplies the computational motivation for the $\mathbb{CP}^2$ example in Section 6.1.

]

Coherent E-Orientations. [**Derek:** Include the following contents:

- For this paragraph, assume that E is a commutative ring spectrum. Explain how compatible E-Thom classes for the stable normal bundles of trajectory spaces can replace stable framings in the Pontryagin–Thom construction. Require multiplicative compatibility along boundary products and coherent compatibility at higher corners. Individual orientability of each moduli space is insufficient.
- Describe the resulting E-module realization $\mathbb{X}_f^E$. When its orientation data are induced by the untwisted Morse framings through the unit $\mathbb{S} \to E$, establish

$$\mathbb{X}_f^E \simeq E \wedge \mathbb{X}_f.$$

  Distinguish these induced orientations from more general orientation choices, whose comparison with the untwisted construction requires additional justification. Relate this dependence to Section 6.2 and explain its relevance to the abstract Floer setting of Section 7. Refer to Cohen's module-spectrum construction (arXiv:0802.2752) and Bonciocat, Section 7 (arXiv:2409.11278v3).

]

6 Examples and Applications

6.1 Attaching Map of $\mathbb{CP}^2$

6.2 Twisted Framings, Thom Spectra, and Atiyah Duality

[**Derek:** Expected at least to achieve here.]

7 Abstraction of the Flow Category

Abstract Framed Flow Categories.

The General Realization Construction.

Data from Floer Theory.

Continuation and Invariance

8 Further Directions

[**Derek:** Introduce the three directions as extensions of the constructions developed above: extracting cohomology operations from Morse moduli spaces, organizing flow categories and their maps into a homotopical framework, and assembling Floer spectra into spectral Fukaya categories. Aim for approximately three to four pages in total.]

8.1 Cohomology Operations via Morse Construction

[**Derek:** Include the following contents:

- Begin with the question of how the information retained by the CJS realization can be expressed through computable cohomology operations. Distinguish the existence of operations on the cohomology of the resulting spectrum from their direct geometric construction using Morse data.
- Introduce gradient trees through the simplest example: a Y-shaped tree with two inputs and one output. Explain how, after suitable choices of Morse functions and perturbations, counting rigid

configurations defines $m_2 : CM^*(f_1) \otimes CM^*(f_2) \to CM^*(f_3)$ and recovers the cup product. Describe how boundary strata of one-dimensional compactified moduli spaces yield the chain-map identity, while parameter families provide homotopies.

- Explain that Steenrod operations require additional symmetry and parameter data, including the interchange of the two inputs. Return to the example of $\mathbb{CP}^2$: for its degree-two generator x, one has $\mathrm{Sq}^2(x) = x^2 \neq 0$, whereas the corresponding operation vanishes on $S^2 \vee S^4$. Explain why this distinction survives stabilization.
- Discuss Cohen–Norbury's *Morse field theory* (arXiv: math/0509681) as an existing geometric construction of operations using metric-graph-flow moduli spaces. Conclude by identifying the comparison with CJS realization as a further learning goal. Keep track of the additional diagonal or coalgebra data needed to describe the cup product at the spectrum level.

]

8.2 The Abouzaid–Blumberg Framework for Floer Homotopy Theory

[**Derek:** Include the following contents:

- Motivate the framework by passing from individual realizations to maps between them and homotopies between those maps. Clearly distinguish the objects of a single flow category from the objects of a category whose objects are themselves flow categories.
- Introduce flow bimodules through continuation trajectories. For flow categories $\mathcal{C}$ and $\mathcal{D}$, explain that a bimodule consists of spaces $\mathcal{B}(x, y)$ whose boundary faces include products of the forms $\mathcal{C}(x, x') \times \mathcal{B}(x', y)$ and $\mathcal{B}(x, y') \times \mathcal{D}(y', y)$. Interpret these as breaking before and after the continuation region. Describe this geometry without reproducing all axioms.
- State the main structural results of Abouzaid–Blumberg, *Foundation of Floer homotopy theory I: Flow categories* (arXiv:2404.03193), Theorem 1.6 and Proposition 1.10: structured flow categories form a stable infinity-category, and their framed version satisfies $\mathrm{Flow}^{\mathrm{fr}} \simeq \mathrm{Sp}$. Explain the associated spectrum through $\mathrm{Map}_{\mathrm{Flow}^{\mathrm{fr}}}(\mathbb{1}, \mathcal{C})$, where $\mathbb{1}$ is the distinguished one-object flow category. Specify that this uses their full framework, beyond the finite Morse setting of the main text.
- Briefly explain the role of structured derived orbifolds when Floer moduli spaces are not smooth manifolds, and mention the Hamiltonian Floer construction in Appendix B. Close by identifying multiplicative structures and multiple-input operations as the additional ingredients needed for the next subsection. Respect the scope described in Section 1.5 of the cited paper.

]

8.3 Spectral Fukaya Categories

[**Derek:** Include the following contents:

- Fix a manageable setting, such as compact exact Lagrangian branes in a Liouville manifold, with the required grading and tangential data. Recall that the ordinary Fukaya category has Lagrangian branes as objects, Floer cochain complexes as morphism objects, and operations defined by holomorphic polygons.
- For each pair L_0, L_1, explain how Floer strips determine a flow category $\mathcal{C}(L_0, L_1)$. Under suitable compatible framing assumptions, its CJS realization provides a candidate morphism spectrum $\mathbb{CF}(L_0, L_1)$. Explain the recovery of ordinary Floer cohomology after passing to $H\mathbb{Z}$-coefficients, specifying the grading convention. Emphasize that this step constructs the morphism objects.
- Explain how triangle moduli spaces provide the geometric input for a composition map $\mathbb{CF}(L_1, L_2) \wedge \mathbb{CF}(L_0, L_1) \to \mathbb{CF}(L_0, L_2)$. Describe why the Pontryagin–Thom constructions must respect boundary strata involving broken strips. This supplies the link between pairwise Floer flow categories and

categorical composition.

- Use the compactified moduli space of quadrilateral domains, an interval, to explain the homotopy between the two ways of composing three morphisms. Then describe higher polygon domains and associahedra as the source of higher coherence. Include units and compatibility with continuation among the required structures.
- Explain the dependence on coefficients and tangential structures. For a commutative ring spectrum R, the relevant input is coherent R-orientation data for the moduli spaces, and composition uses $\wedge_R$. Refer to Porcelli–Smith, *Spectral Floer theory and tangential structures* (arXiv:2411.03257).
- Conclude with a precise account of existing results: Asplund–Deshmukh–Pieloch, *Spectral equivalence of nearby Lagrangians* (arXiv:2411.08841), Theorem 1.2, constructs a wrapped Donaldson–Fukaya category enriched over the homotopy category of R-modules. Distinguish this from retaining all A_∞-coherence. As a concrete next example, propose comparing the Morse construction with Floer theory for Q and $\mathrm{graph}(\varepsilon \mathrm{d}f)$ in T^*Q, including compatibility with products.

]

Appendices

A Simplicial Spaces and Classifying Spaces

B Elementary Facts in Stable Homotopy Theory

C Hamilton and Lagrangian Floer Theory

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